Ratios, Proportions & Percentages

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

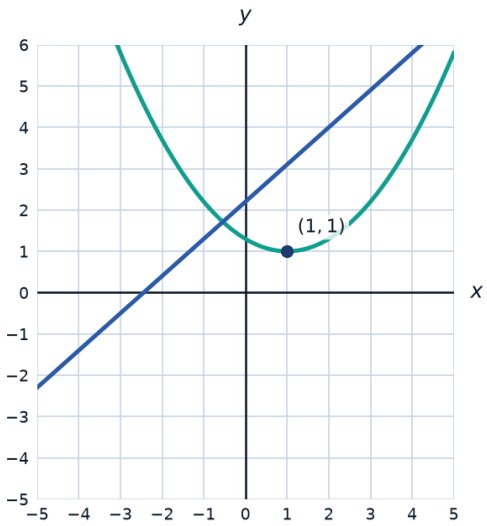
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying ratios

Simplify the ratio $45 : 60 : 75$.

2 [Foundational] • Target: Simplifying ratios

Express "12 boys to 18 girls" as a ratio in simplest form.



3 [Foundational] • Target: Sharing a quantity in a given ratio

Share 150 in the ratio $2 : 3$.

4 [Foundational] • Target: Sharing a quantity in a given ratio

Share 220 among three people in the ratio $3 : 4 : 4$.

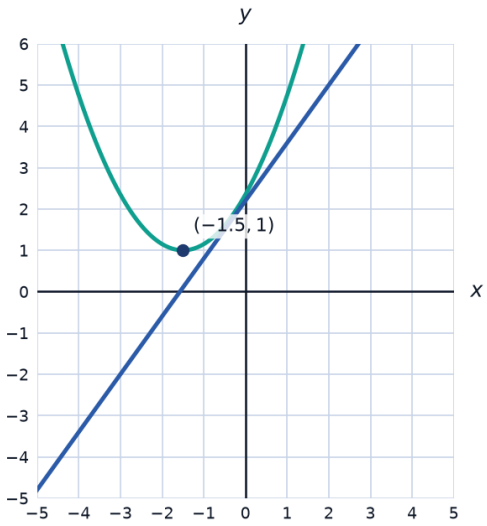
5 [Foundational] • Target: Sharing a quantity in a given ratio

A prize of 900 riyals is shared between Sara and Layla in the ratio $5 : 4$. Find each share.

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6 [Foundational] • Target: Sharing a quantity in a given ratio

Divide 84 kg of flour in the ratio $1:2:4$.



7 [Foundational] • Target: Equivalent ratios and proportions

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many meters in reality?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write $\frac{3}{8}$ as a percent.

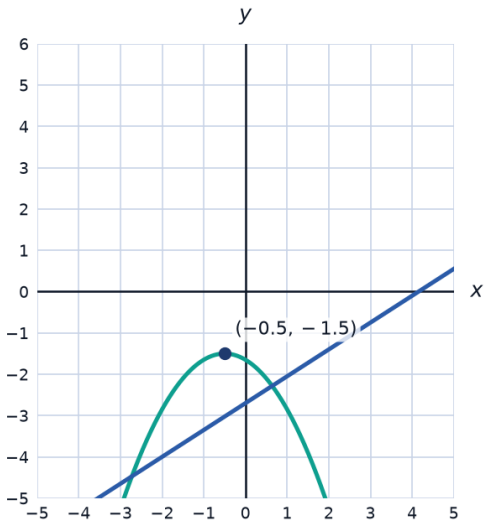
9 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write 0.45 as a fraction in lowest terms.

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10 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write $\frac{7}{40}$ as a decimal.



11 [Intermediate] • Target: Percentage increase and decrease

A salary of 4,000 riyals increases by 6%. Find the new salary.

12 [Intermediate] • Target: Percentage increase and decrease

A population of 2,500 decreases by 12%. Find the new population.

13 [Intermediate] • Target: Finding percentage change and original values

A price rises from 80 to 92. Find the percentage increase.

14 [Intermediate] • Target: Finding percentage change and original values

A quantity falls from 250 to 200. Find the percentage decrease.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

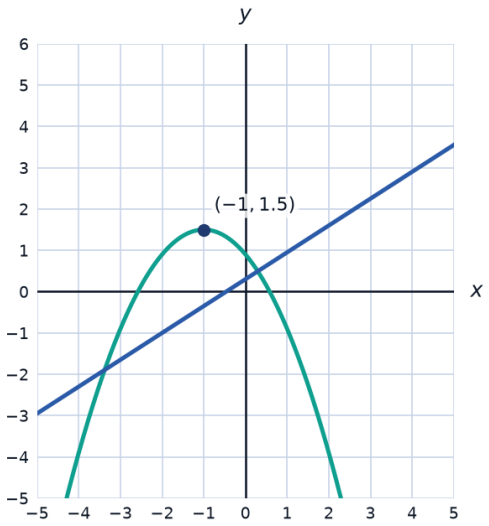
15 [Challenging] • Target: Finding percentage change and original values • Cross-Lesson: 5.01 Quantitative Comparisons

After a 20% discount, a jacket costs 96 riyals. Find the original price.

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16 [Challenging] • Target: Finding percentage change and original values • Cross-Lesson: 6.01 Sequences & Patterns

After a 15% increase, a value becomes 138. Find the original value.



17 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 7.01 Data Analysis & Probability

A price of 100 rises 10%, then falls 10%. Find the final price.

18 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 8.01 Word Problems & Reasoning

A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.

19 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 1.01 Arithmetic & Number Sense

A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.

20 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 3.01 Algebra Essentials

A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Simplifying & writing ratios

A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.

a. $1:125$

b. $250:2$

c. $125:1$

d. $2:250$

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22 [Foundational] • Target: MCQ — Simplifying & writing ratios

Express $0.75 : 2$ as a ratio of whole numbers in simplest form.

a. $3 : 4$

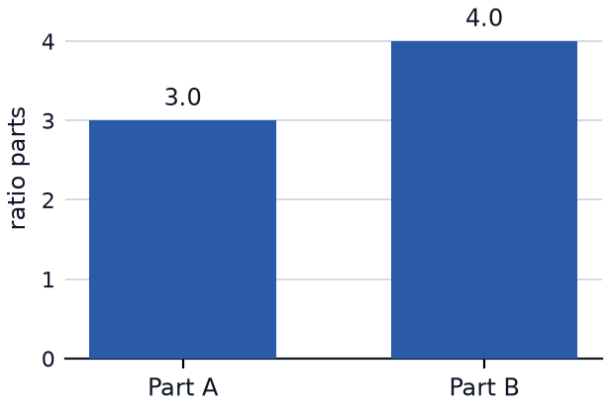
b. $3 : 8$

c. $75 : 2$

d. $75 : 200$

23 [Foundational] • Target: MCQ — Sharing in a given ratio

The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.



a. 36

b. 21

c. 48

d. 63

24 [Foundational] • Target: MCQ — Sharing in a given ratio

Share 260 SAR in the ratio $2 : 3 : 5$. Find the smallest share.

a. 78 SAR

b. 52 SAR

c. 130 SAR

d. 65 SAR

25 [Foundational] • Target: MCQ — Sharing in a given ratio

Two business partners share a profit of $18{,}000$ SAR in the ratio $5 : 7$. Find the larger share.

a. $10{,}500$ SAR

b. $9{,}000$ SAR

c. $7{,}500$ SAR

d. $12{,}857$ SAR

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26 [Foundational] • Target: MCQ — Sharing in a given ratio

A line of length 36 cm is divided in the ratio $4:5$. Find the length of the shorter piece.

a. 16

b. 4

c. 9

d. 20

27 [Foundational] • Target: MCQ — Sharing in a given ratio

A class of 40 students has boys and girls in the ratio $3:5$. How many boys are there?

a. 25

b. 8

c. 24

d. 15

28 [Foundational] • Target: MCQ — Equivalent ratios & scale

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many km in reality?

a. 15 km

b. 1.5 km

c. 150 km

d. 0.15 km

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Equivalent ratios & scale

Which ratio is equivalent to $4:9$?

a. $9:4$

b. $20:45$

c. $8:16$

d. $16:37$

30 [Intermediate] • Target: MCQ — Equivalent ratios & scale

In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?

a. 4.8

b. 24

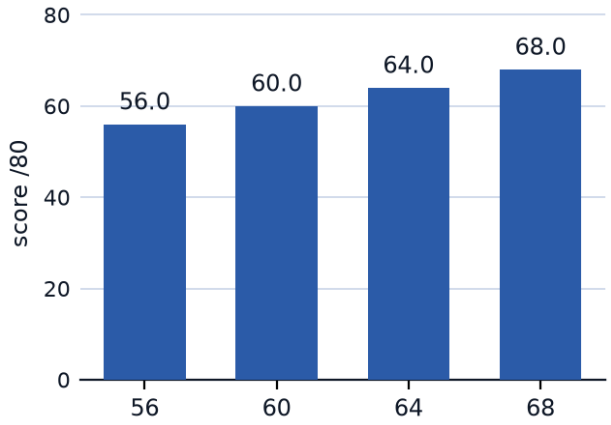
c. 7.2

d. 12

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31 [Intermediate] • Target: MCQ — Percent of a number

The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?



a. 68

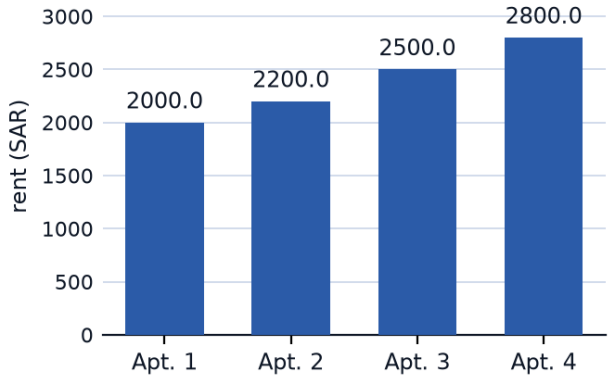
b. 64

c. 60

d. 56

32 [Intermediate] • Target: MCQ — Percent of a number

The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?



a. Apt. 2

b. Apt. 1

c. Apt. 3

d. Apt. 4

33 [Intermediate] • Target: MCQ — Percent of a number

150% of a number is 300. Find the number.

a. 250

b. 450

c. 200

d. 150

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34 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Write 85% as a fraction in simplest form.

a. $\frac{17}{20}$

b. $\frac{8}{5}$

c. $\frac{17}{25}$

d. $\frac{85}{100}$

35 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Which of these is largest: $\frac{3}{8}$, 0.4, or 37%?

a. 0.4

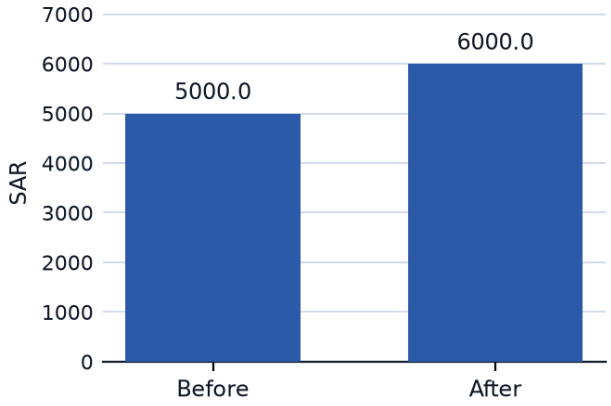
b. They are all equal

c. $\frac{3}{8}$

d. 37%

36 [Intermediate] • Target: MCQ — Percentage increase & decrease

The bar chart shows a salary before and after a raise. What percentage increase does it represent?



a. 15%

b. 500%

c. 20%

d. 25%

37 [Intermediate] • Target: MCQ — Percentage increase & decrease

A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 SAR

b. 144 SAR

c. 100 SAR

d. 140 SAR

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Percentage increase & decrease • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the percentage change from 40 to 34.

a. 17.6% decrease

b. 15% increase

c. 6% decrease

d. 15% decrease

39 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 1.01 Arithmetic & Number Sense

After a 15% increase, a price is 92 SAR. Find the original price.

a. 80 SAR

b. 78.20 SAR

c. 88 SAR

d. 107 SAR

40 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 3.01 Algebra Essentials

After a 30% discount, an item costs 56 SAR. Find the original price.

a. 72.80 SAR

b. 80 SAR

c. 86 SAR

d. 74.67 SAR

41 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 4.01 Geometry & Measurement

A population grew by 8% to reach 27,000. Find the population before the growth.

a. 25,000

b. 26,000

c. 29,160

d. 24,840

42 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 5.01 Quantitative Comparisons

After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.

a. 158.70 SAR

b. 120 SAR

c. 123 SAR

d. 117.30 SAR

43 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 6.01 Sequences & Patterns

A price of 200 rises 10%, then rises another 10%. Find the final price.

a. 220

b. 240

c. 244

d. 242

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44 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 7.01 Data Analysis & Probability

A price of 500 falls 20%, then rises 20%. Find the final price.

a. 460

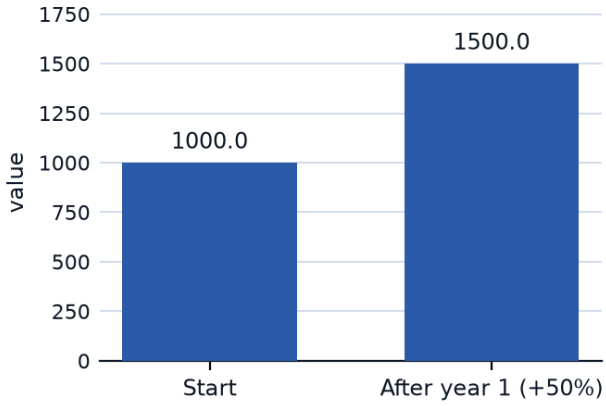
b. 480

c. 500

d. 520

45 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 8.01 Word Problems & Reasoning

The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.



a. 500

b. 900

c. 1,000

d. 750

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